



EE/CE/RE 3247 – EMBEDDED SYSTEMS

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SMART CLASSROOM AUTOMATION SYSTEM

*Controls Lights, Fan, and Projector Based on Occupancy Using Sensors,
Relays, and a Microcontroller to Reduce Power Consumption*

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CHAPTER 1: INTRODUCTION

1.1 Abstract

This report presents the design, development, and testing of a Smart Classroom Automation System built to reduce unnecessary energy consumption in educational facilities. The system uses two passive infrared (PIR) sensors — one at the entry point and one mounted on the ceiling — to detect occupancy and automatically control classroom appliances including lights, a fan, and a projector through a relay module. An ATmega328P microcontroller serves as the processing unit, executing a non-blocking state machine that manages system states: STANDBY and ACTIVE. A 16x2 I2C LCD provides real-time status feedback. When motion is detected, all appliances are switched on and a 2-second countdown timer is initiated. Continued motion detection resets the timer, keeping appliances on for as long as the room is occupied. Upon expiry of the timer without motion, the system reverts to STANDBY and all appliances are switched off. The DC motor used to drive the fan was found to draw significant current and required an independent power supply to prevent interference with the main circuit. Testing confirmed reliable PIR sensitivity, responsive relay switching, and stable system operation. The system demonstrates a practical, low-cost approach to smart energy management in classroom environments.

1.2 Background of the Study

Energy consumption in educational institutions represents a significant operational cost, particularly when electrical appliances such as lights, fans, and projectors are left running in unoccupied rooms. In many Ghanaian universities and schools, manual control of these appliances is the norm, leading to energy waste due to human oversight or negligence. Automating appliance control based on room occupancy offers a straightforward and effective solution to this problem.

This project proposes and implements a Smart Classroom Automation System that leverages occupancy sensing to automatically manage classroom electrical appliances. The system is built around the ATmega328P microcontroller and uses PIR sensors to

detect motion, relay modules to switch appliances, and an LCD display to provide real-time status feedback. The core objective is to ensure that energy is consumed only when the classroom is in use, thereby reducing waste and promoting sustainable energy practices in educational settings.

The system operates on a timer-based logic: when motion is detected, appliances are activated and a countdown timer is reset. As long as occupancy is sensed, the timer continues to reset, keeping appliances on. When no motion is detected for the duration of the timeout period, the system automatically switches off all appliances. This design ensures that the system is responsive to actual room usage without requiring any manual intervention.

1.3 Problem Statement

Educational institutions in Ghana and across the developing world face persistent challenges with unnecessary energy consumption resulting from the manual control of classroom appliances. Lights, fans, and projectors are routinely left running in unoccupied classrooms due to human oversight, forgetfulness, or lack of accountability, leading to avoidable electricity costs and increased carbon footprint. In institutions where energy budgets are already constrained, this inefficiency represents a significant and addressable problem.

Existing manual switching systems place full responsibility on occupants to manage appliance states upon entering and leaving a room. This approach is unreliable in practice, particularly in high-traffic educational environments where classrooms are used by multiple groups throughout the day. The absence of an automated, occupancy-aware control mechanism means that energy is wasted during breaks, after lectures, and during unscheduled periods when rooms are empty but appliances remain active.

This project addresses this problem by designing and implementing an embedded system that automatically detects classroom occupancy using passive infrared sensors and controls lights, a fan, and a projector accordingly, eliminating the dependence on manual switching and ensuring that energy is consumed only when the room is in active use.

1.4 Project Objectives

- i. To design and implement an occupancy-based classroom automation system using PIR sensors and a microcontroller.
- ii. To automatically control lights, a fan, and a projector relay based on detected motion.
- iii. To provide real-time system status feedback via an I2C LCD display.
- iv. To reduce unnecessary energy consumption in classroom environments.
- v. To demonstrate a reliable embedded systems solution using the ATmega328P microcontroller.

The remainder of this report details the project's execution, findings, and conclusions.

CHAPTER 2: LITERATURE REVIEW

Occupancy-based automation systems have been a growing area of research and development in the field of embedded systems and smart building technology. The fundamental concept relies on the ability to detect human presence and use that information to trigger automated responses in the built environment.

2.1 PIR-Based Occupancy Detection

Passive Infrared (PIR) sensors are widely used in occupancy detection systems due to their low cost, ease of integration, and reliable performance. PIR sensors detect changes in infrared radiation caused by the movement of warm bodies, making them well-suited for detecting human presence in a room. Studies have shown that PIR sensors achieve high detection accuracy in controlled indoor environments, though they are sensitive to environmental factors such as HVAC airflow and direct sunlight [1].

In classroom settings, PIR sensors have been successfully deployed in energy management systems. Research by Nguyen and Aiello [2] demonstrated that occupancy-driven control of lighting systems in office buildings can achieve energy savings of up to 30% compared to manual switching. Similar results have been reported in educational buildings, where the combination of occupancy sensors with automated lighting control has proven effective.

2.2 Relay-Based Appliance Control

Relay modules serve as the interface between low-voltage microcontroller outputs and high-voltage appliance circuits. In embedded systems applications, relay modules allow microcontrollers to switch AC or DC loads without direct electrical connection, providing both control flexibility and circuit protection [3]. The use of electromechanical relays in smart control systems is well-documented, with applications ranging from industrial automation to home energy management.

A key challenge with relay-based motor control is electromagnetic interference (EMI) generated by inductive loads such as DC motors. Back-electromotive force (back-EMF) produced when motors switch off can cause voltage spikes that disrupt sensitive components such as microcontrollers and LCD displays. The standard mitigation technique involves placing a flyback diode across the motor terminals to suppress these spikes [4].

2.3 Microcontroller-Based Automation

The ATmega328P, the microcontroller at the heart of the Arduino Uno platform, is one of the most widely used microcontrollers in embedded systems education and prototyping. It offers a rich set of digital I/O pins, hardware I2C and SPI support, and a well-established development ecosystem. Its use in classroom automation projects has been extensively documented in the literature and in educational project repositories [5].

Non-blocking programming techniques, such as the use of `millis()` for timing rather than `delay()`, are considered best practice in embedded systems design. This approach allows the microcontroller to continue executing other tasks while waiting for timer events, resulting in more responsive and reliable system behavior [6].

2.4 I2C LCD Displays in Embedded Systems

The I2C (Inter-Integrated Circuit) protocol is widely used for connecting peripheral devices such as LCD displays to microcontrollers using only two signal lines: SDA (data) and SCL (clock). The `LiquidCrystal_I2C` library provides a simple interface for driving 16x2 LCD modules via I2C. A known requirement for I2C communication on bare microcontroller implementations is the inclusion of pull-up resistors on the SDA and SCL lines, as these lines are open-drain by design and require external pull-ups to achieve valid logic high levels [7].

Chapter 3: Methodology

The development of the Smart Classroom Automation System followed a systematic hardware-first approach, beginning with component selection and circuit design, followed by firmware development and iterative testing.

3.1 System Architecture

The system is structured around three functional layers: sensing, processing, and actuation. The sensing layer consists of two PIR sensors that continuously monitor for motion within the classroom. The processing layer is handled by the ATmega328P microcontroller, which implements the system logic and state machine. The actuation layer comprises a three-channel relay module that switches the classroom appliances on and off based on commands from the microcontroller. An I2C LCD display provides a user-facing status interface.

3.2 Component Selection

The following components were selected for the implementation. Each was chosen based on suitability for the system requirements, availability, and compatibility with the ATmega328P microcontroller.

3.2.1 ATmega328P Microcontroller

The ATmega328P is an 8-bit AVR microcontroller manufactured by Microchip Technology. It operates at up to 16 MHz using an external crystal oscillator and runs on a 5V supply. It features 23 programmable I/O pins, 32KB of flash memory for program storage, and hardware support for I2C, SPI, and UART communication protocols. In this system it serves as the central processing unit, executing the occupancy detection logic, managing system states, controlling relay outputs, and driving the LCD display.

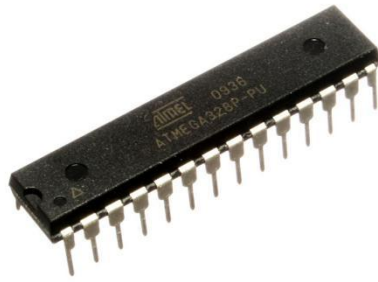


Figure 3.2.1 Image of ATmega328P Microcontroller [I2]

3.2.2 16MHz Crystal Oscillator

The 16MHz crystal oscillator provides the external clock signal that drives the ATmega328P at its full operating speed. It is connected across the XTAL1 and XTAL2 pins of the microcontroller, with a 22pF ceramic capacitor on each leg connected to ground. The crystal ensures accurate and stable timing for all time-dependent operations in the firmware, including the `millis()`-based countdown timer, I2C communication, and relay switching logic. Without an external crystal, the ATmega328P falls back to its internal 8MHz RC oscillator, which is less accurate and would cause timing errors and I2C communication failures.



Figure 3.2.2. Image of the 16Mhz Crystal Oscillator [I6]

3.2.3 HC-SR501 PIR Sensor

The HC-SR501 is a passive infrared motion sensor that detects changes in infrared radiation caused by the movement of warm bodies within its field of view. It operates on 5V, has an adjustable detection range of up to 7 metres, and outputs a digital HIGH signal when motion is detected. Two units are used in this system — one mounted at the classroom entry point to detect persons entering and exiting, and one mounted on the ceiling to detect in-room motion and prevent false STANDBY transitions while the room is occupied.



Figure 3.2.3 Image of HC-SR501 PIR Sensor [15]

3.2.4 4-Channel Relay Module

Though our system utilizes a 4-Channel relay module, only 3 of those were utilized. The relay module provides electrical isolation between the low-voltage microcontroller circuit and the higher-voltage appliance loads. Each channel contains an electromechanical relay rated for switching loads up to 10A at 250V AC or 30V DC. The module accepts 5V control signals directly from the microcontroller's digital output pins. Three channels are used to independently switch the classroom lights, projector, and fan.

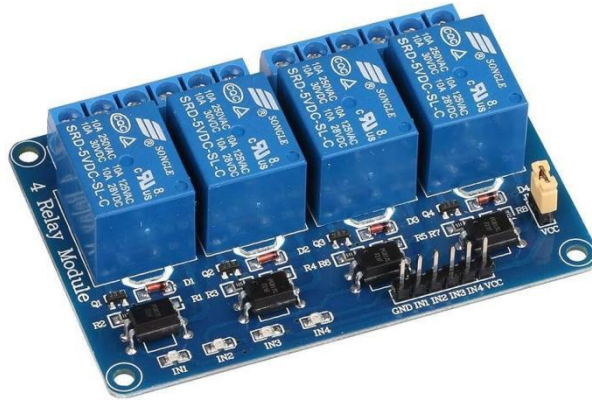


Figure 3.2.4 Image of 4-Channel Relay Module [I11]

3.2.5 16x2 I2C LCD Display

The 16x2 liquid crystal display provides two rows of sixteen characters each for real-time system status feedback. It is interfaced via the I2C protocol using an onboard PCF8574 I/O expander, which reduces the number of required connections to just two signal lines — SDA and SCL — in addition to power and ground. The module operates at 5V and uses the I2C address 0x27. It displays the current system state (STANDBY or ACTIVE), the source of motion detection, and a live countdown timer.



Figure 3.2.5 Image of 16x2 I2C LCD [I12]

3.2.6 Small Brushed DC Motor

A small brushed DC motor is used to drive the classroom fan blade. Brushed DC motors operate by passing current through carbon brushes that make contact with a rotating commutator, generating rotational motion through electromagnetic interaction. They are simple to control, require no specialised driver circuitry for basic on/off operation, and are well suited for low-cost embedded systems applications. The motor operates within a voltage range of 3V to 6V. Due to its high current draw on startup and the back-EMF it generates when switching off, the motor is powered from a dedicated external supply rather than the shared 5V rail, and a 1N4007 flyback diode is placed across its terminals to suppress voltage transients and protect the rest of the circuit.



Figure 3.2.6 Image of Small Brushed DC Motor [11]

3.2.7 LEDs (x2)

Two light-emitting diodes are used in the prototype to simulate the classroom light and projector loads. They serve as visual indicators of relay switching behaviour during testing and demonstration, representing the actual high-power appliances that would be connected in a full installation.



Figure 3.2.7 Image of LED [14]

3.2.8 1N4007 Flyback Diode

The 1N4007 is a general-purpose rectifier diode rated for 1A forward current and 1000V reverse voltage. It is placed in reverse bias across the DC motor terminals — cathode to motor positive, anode to motor negative — to clamp and dissipate the voltage spikes generated by the motor's inductive winding when the relay switches it off. Without this diode, back-EMF transients can propagate through the ground line and disrupt other components in the circuit.



Figure 3.2.8 Image of 1N4007 Flyback Diode [17]

3.2.9 4.7k Ω Pull-up Resistors (x2)

Two 4.7k Ω resistors are connected between the I2C SDA and SCL lines and the 5V supply rail. The I2C protocol uses open-drain signalling, meaning the bus lines are

only actively pulled low by devices and rely on external resistors to return to a logic high state. On a bare ATmega328P, unlike the Arduino Uno, these pull-ups are not built into the board and must be added externally. Without them, SDA and SCL remain near 0V and I2C communication fails entirely.



Figure 3.2.9 Image of Pull-up Resistor [18]

3.2.10 22pF Crystal Capacitors (x2)

Two 22 picofarad ceramic capacitors are connected between each leg of the 16MHz crystal oscillator and ground. These load capacitors are required to stabilise the crystal's oscillation frequency and ensure the ATmega328P clocks at the correct 16MHz rate. Incorrect or missing capacitors cause the microcontroller to run at the wrong speed, which directly affects time-sensitive operations such as I2C communication and `millis()`-based timing.



Figure 3.2.10 Image of 22pF Crystal Capacitor [110]

3.3 Circuit Design

The circuit was designed with the ATmega328P as the central node. The two PIR sensors connect their signal output pins to digital pins 4 (pin 6) and 2 (pin 4) of the microcontroller respectively. The three relay module input pins (IN1, IN2, IN3) connect to digital pins 5, 6, and 7 (pins 11, 12, and 13 respectively) to control the lights, projector, and fan respectively. The relay module's VCC and GND connect to the ATmega328P's 5V and GND rails.

The I2C LCD module connects via SDA to pin 27 (PC4) and SCL to pin 28 (PC5) of the ATmega328P. Two 4.7k Ω pull-up resistors are placed between SDA and 5V, and between SCL and 5V, as required for I2C operation on a bare ATmega328P (unlike the Arduino Uno, which includes built-in pull-ups).

The DC motor is powered by a dedicated external 5–6V supply to isolate it from the main circuit. The motor connects through one of the relay channels, with a 1N4007 flyback diode placed across its terminals (cathode to motor positive, anode to motor negative) to suppress back-EMF spikes. The negative terminal of the external motor supply is connected to the designated GND on the PCB to establish a common ground between the two circuits.

3.4 Block Diagram

Below shows the block diagram of the system:

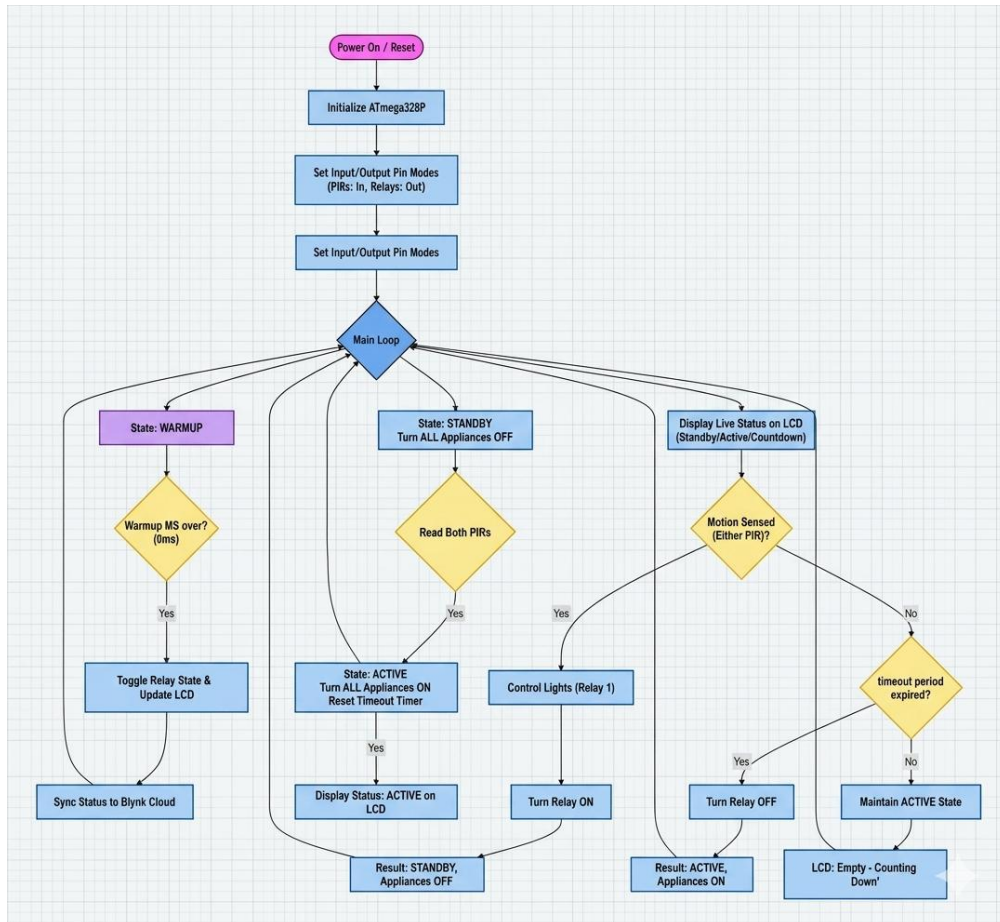


Figure 3.4.1 Block Diagram

3.5 Schematic/Circuit Diagram

Below are the hand-drawn schematic and software circuit diagrams repectively. The circuit diagram was designed with the **Cirkit Designer website**.

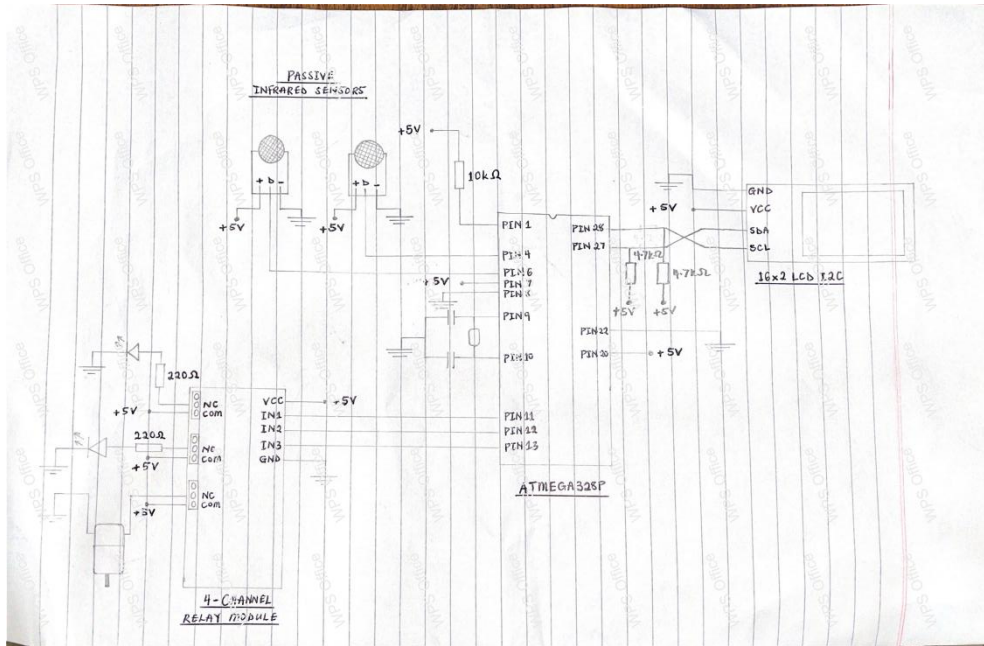


Figure 3.4.1 Schematic Diagram

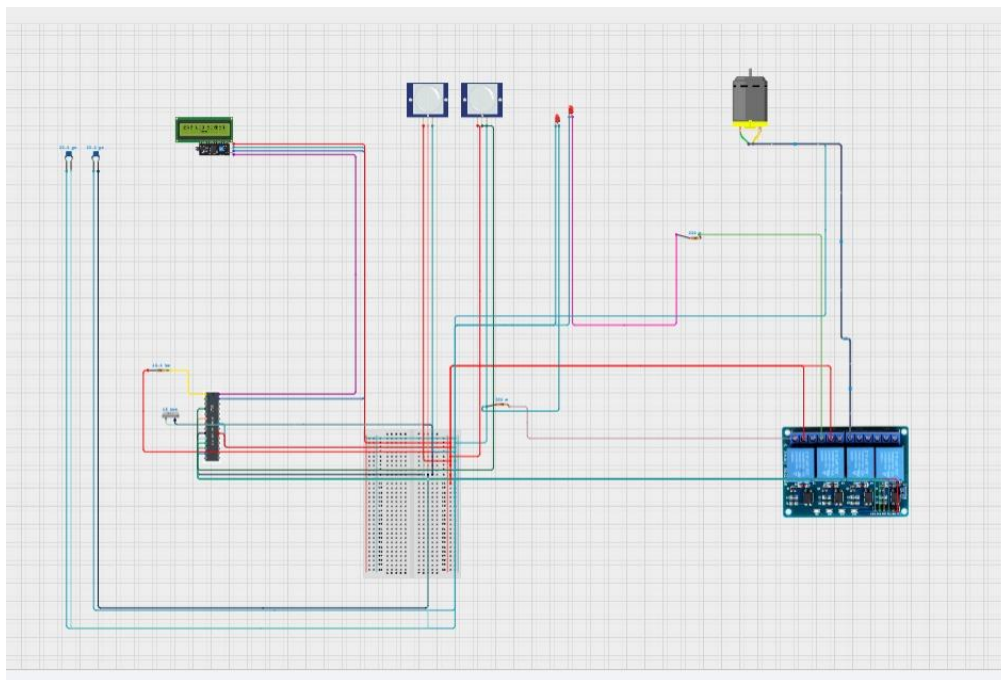


Figure 3.4.2 Circuit Diagram

3.6 Firmware Design

The firmware was written in C++ using the Arduino framework and implements a non-blocking state machine with three states: **WARMUP**, **STANDBY**, and **ACTIVE**. All timing operations use `millis()` to avoid blocking the main execution loop.

The system initializes in the WARMUP state. With WARMUP_MS set to 0, the system transitions immediately to STANDBY on the first loop iteration, displaying 'Status: STANDBY' and 'Room Empty' on the LCD. Both relay outputs and the motor are explicitly set to LOW during initialization to ensure all appliances start in the off state.

In the main loop, both PIR sensors are read on every iteration. If either sensor detects motion (HIGH output), the timerStart is reset to the current millis() value, and if the system is not already in ACTIVE state, it transitions to ACTIVE and switches all relay outputs HIGH, turning on all appliances. The LCD updates to show 'Status: ACTIVE'.

While in ACTIVE state, the system continuously checks whether the elapsed time since the last motion detection exceeds TIMEOUT_MS (2000 ms). If no motion is detected within this window, the system transitions back to STANDBY and switches all outputs LOW. If motion continues to be detected, the timer resets on every iteration, ensuring appliances remain on throughout the period of occupancy. The LCD provides a live countdown display showing remaining time in seconds and tenths of seconds, and updates to show which sensor triggered the reset when motion is detected.

3.7 Program Code

The following excerpt shows the entire project program:

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

// --- LCD Configuration ---
LiquidCrystal_I2C lcd(0x27, 16, 2);

// --- Pin Definitions ---
const int pirEntryPin = 4; // Entry/exit PIR
const int pirCeilingPin = 2; // Ceiling PIR
const int relayPin1 = 5; // IN1 - Lights
const int relayPin2 = 6; // IN2 - Projector
const int relayPin3 = 7; // IN3 - Fan

// --- Timer Settings ---
const unsigned long TIMEOUT_MS = 2000UL; // 2 second off-delay
const unsigned long WARMUP_MS = 0UL; // No PIR warmup

// --- State ---
enum State { WARMUP, STANDBY, ACTIVE };
State currentState = WARMUP;
```

```

unsigned long timerStart = 0;
unsigned long lastLCDUpdate = 0;
unsigned long warmupStart = 0;

void setOutputs(bool on) {
  digitalWrite(relayPin1, on ? HIGH: LOW);
  digitalWrite(relayPin2, on ? HIGH: LOW);
  digitalWrite(relayPin3, on ? HIGH: LOW);
}

void setup() {
  pinMode(relayPin1, OUTPUT);
  pinMode(relayPin2, OUTPUT);
  pinMode(relayPin3, OUTPUT);
  pinMode(pirEntryPin, INPUT);
  pinMode(pirCeilingPin, INPUT);
  setOutputs(false);

  // Explicitly start all relays OFF
  digitalWrite(relayPin1, LOW);
  digitalWrite(relayPin2, LOW);
  digitalWrite(relayPin3, LOW);

  delay(500); // give LCD time to power up
  Wire.begin();
  lcd.init();
  lcd.backlight();
  lcd.setCursor(0, 0);
  lcd.print("PIR Warmup ");
  lcd.setCursor(0, 1);
  lcd.print("Please wait... ");

  warmupStart = millis();
}

void loop() {
  unsigned long now = millis();

  // --- WARMUP phase (non-blocking) ---
  if (currentState == WARMUP) {
    if (now - lastLCDUpdate >= 1000) {
      unsigned long remaining = WARMUP_MS - (now - warmupStart);
      unsigned int secsLeft = remaining / 1000;

      lcd.setCursor(0, 1);
      lcd.print("Ready in: ");
      lcd.print(secsLeft);
      lcd.print("s ");

      lastLCDUpdate = now;
    }

    if (now - warmupStart >= WARMUP_MS) {
      currentState = STANDBY;
      setOutputs(false);
      lcd.setCursor(0, 0);
      lcd.print("Status: STANDBY ");
      lcd.setCursor(0, 1);
      lcd.print("Room Empty ");
    }
  }
}

```

```

    }
    return;
}

// --- Read both PIRs ---
bool entryMotion  = digitalRead(pirEntryPin) == HIGH;
bool ceilingMotion = digitalRead(pirCeilingPin) == HIGH;
bool motionDetected = entryMotion || ceilingMotion;

// --- Either PIR triggered: reset timer ---
if (motionDetected) {
    timerStart = now;

    if (currentState != ACTIVE) {
        currentState = ACTIVE;
        setOutputs(true);
        lcd.setCursor(0, 0);
        lcd.print("Status: ACTIVE ");
        lastLCDUpdate = 0; // force immediate LCD update
    }
}

// --- Active: countdown or timeout ---
if (currentState == ACTIVE) {
    unsigned long elapsed = now - timerStart;
    unsigned long remaining = (elapsed >= TIMEOUT_MS) ? 0 : TIMEOUT_MS - elapsed;

    if (elapsed >= TIMEOUT_MS) {
        currentState = STANDBY;
        setOutputs(false);
        lcd.setCursor(0, 0);
        lcd.print("Status: STANDBY ");
        lcd.setCursor(0, 1);
        lcd.print("Room Empty   ");
    }
    else if (now - lastLCDUpdate >= 100) {
        unsigned int secsLeft = remaining / 1000;
        unsigned int msLeft = (remaining % 1000) / 100;

        lcd.setCursor(0, 1);
        if (motionDetected) {
            if (entryMotion && ceilingMotion) {
                lcd.print("Both: Reset! ");
            }
            else if (entryMotion) {
                lcd.print("Entry: Reset! ");
            }
            else {
                lcd.print("Ceiling: Reset! ");
            }
        }
        else {
            lcd.print("Off in: ");
            lcd.print(secsLeft);
            lcd.print(".");
            lcd.print(msLeft);
            lcd.print("s ");
        }
    }

    lastLCDUpdate = now;
}
}
}

```

Chapter 4: Results

The system was assembled and tested on a breadboard prototype before migration to a PCB-based implementation with the ATmega328P. The following results were observed during testing.

4.1 Occupancy Detection and Timer Behaviour

Both PIR sensors demonstrated high sensitivity in detecting motion within the classroom environment. The entry PIR reliably detected persons entering and exiting the classroom, while the ceiling PIR detected smaller in-room movements. Due to the high sensitivity of the sensors, even minor motion within the classroom triggered a timer reset, ensuring that appliances were not inadvertently switched off while occupants were present. This behaviour confirmed the practical suitability of the two-sensor configuration for classroom use.

The 2-second timeout ($\text{TIMEOUT_MS} = 2000 \text{ ms}$) was observed to function correctly. The LCD displayed a live countdown from 2.0s to 0.0s in 100 ms increments when no motion was detected. Upon any motion event, the timer reset immediately and the LCD updated to display the source of the trigger (entry sensor, ceiling sensor, or both).

4.2 Relay and Appliance Switching

All three relay channels switched reliably in response to microcontroller output signals. The LEDs representing the classroom lights and projector responded instantaneously to relay state changes. The DC motor fan activated correctly through its dedicated relay channel.

4.3 Power and Current Observations

A total system current draw of approximately 0.22A was measured on the main 5V supply during normal operation (excluding the DC motor). This reading confirmed that the main supply was adequately sized for the ATmega328P, PIR sensors, relay module, and LCD combined.

During initial testing with the DC motor connected to the main 5V rail, significant interference was observed: the LCD display went blank, PIR sensors behaved erratically, and the relay module exhibited unstable switching. This was attributed to the motor's high current draw and back-EMF spikes loading the shared power rail. The issue was fully resolved by powering the motor from a dedicated external supply and adding a 1N4007 flyback diode across the motor terminals. After this modification, all components operated stably and without interference.

4.4 LCD Display

On the Arduino Uno platform, the LCD displayed correctly using the I2C address 0x27, confirmed via an I2C scanner sketch. On the bare ATmega328P, the LCD backlight powered on but displayed no text until 4.7k Ω pull-up resistors were added to the SDA and SCL lines. Without pull-up resistors, measured voltage on the I2C lines was approximately 0.3–0.5V, confirming that the lines were being held low and I2C communication was failing. After adding the pull-up resistors on a soldered PCB (breadboard connections proved unreliable), the I2C line voltages returned to approximately 5V and the LCD displayed correctly.

Chapter 5: Discussion

The Smart Classroom Automation System successfully achieved its primary objective of automating appliance control based on occupancy sensing. The non-blocking state machine approach proved effective, allowing the system to simultaneously manage PIR polling, relay switching, timer logic, and LCD updates without any observable lag or missed events.

The decision to use two PIR sensors rather than one significantly improved system reliability. A single entry-point sensor would be insufficient for detecting occupants already seated in the room, potentially causing false STANDBY transitions. The ceiling-mounted PIR provides redundant coverage of the room interior, and the OR logic between the two sensors ensures that either sensor alone is sufficient to keep the system in ACTIVE state. The high sensitivity of the HC-SR501 sensors was particularly advantageous in this regard, as minor movements such as writing or turning a page were sufficient to reset the timer.

The motor interference issue highlighted a fundamental principle in mixed-load embedded circuit design: inductive loads should never share a power rail with sensitive digital components. The current spikes generated by the DC motor on startup and the back-EMF transients on shutdown were sufficient to corrupt I2C communication and disrupt PIR signal readings. The solution — a dedicated power supply for the motor, a flyback diode, and a common ground connection — is consistent with best practices documented in the embedded systems literature.

The I2C LCD troubleshooting process revealed a common pitfall when migrating from Arduino Uno to bare ATmega328P: the Uno includes built-in I2C pull-up resistors, while the bare chip does not. The breadboard implementation proved too unreliable for stable I2C communication, with measured SDA/SCL voltages as low as 0.3V due to poor contact resistance. Migration to a soldered PCB resolved this completely, underscoring the importance of reliable physical connections in embedded systems prototyping.

The 2-second timeout was chosen as a balance between responsiveness and stability. A shorter timeout risks nuisance switching in situations where occupants are briefly still, while a longer timeout delays the system's response to a room becoming empty. For a production deployment, this value could be made configurable, for example via

a potentiometer-adjusted input or a stored setting, to allow adjustment to different classroom usage patterns.

Chapter 6: Recommendations

Based on the design, implementation, and testing of the system, the following recommendations are made for future development and deployment:

- i. Increase timeout configurability: The 2-second off-delay should be made adjustable to suit different classroom environments. A hardware potentiometer or a simple configuration menu via the LCD could allow facility managers to tune the timeout without reprogramming the microcontroller.
- ii. Add a manual override switch: A physical toggle switch should be added to allow occupants to manually keep appliances on regardless of PIR state, for example during extended periods of stillness such as examinations.
- iii. Implement individual appliance control: Currently, all three appliances switch together. A more refined system could control lights, fan, and projector independently based on time of day, ambient light levels, or separate occupancy zones.
- iv. Migrate to a PCB-based design: The breadboard prototype is unsuitable for permanent installation. A custom PCB with soldered connections ensures reliable I2C communication and long-term mechanical stability, as demonstrated during testing.
- v. Add a light-dependent resistor (LDR): An LDR sensor would allow the system to suppress the lights relay during daytime when natural light is sufficient, further reducing energy consumption.
- vi. Incorporate WiFi connectivity: Adding an ESP8266 or ESP32 module would enable remote monitoring of classroom occupancy and energy usage, and allow administrators to receive notifications or override the system remotely.
- vii. Use a motor driver for fan speed control: Replacing the relay-switched DC motor with a PWM-controlled motor driver (such as the L298N) would allow variable fan speed based on the number of occupants or ambient temperature, improving comfort and energy efficiency.
- viii. Conduct extended field testing: The system should be deployed in an actual classroom environment over an extended period to validate performance under

real-world conditions including varying ambient temperatures, different lighting conditions, and varying occupancy patterns.

Chapter 7: References

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Appendix A: Project Photo Documentation

This appendix contains photographs taken throughout the various phases of the Smart Classroom Automation System project, including component assembly, breadboard prototyping, circuit testing, and the final assembled system in operation.

A.1 Final Assembled System

The following photographs show the completed Smart Classroom Automation System, including the ATmega328P on PCB, relay module, LCD display, PIR sensors, DC motor fan, and associated wiring.



Figure A.1: Final assembled Smart Classroom Automation System



Figure A.2: Final assembled system — alternate view

A.2 System in Operation

These photographs capture the system actively running, showing the LCD display in STANDBY and ACTIVE states, relay switching, and the DC motor fan in operation.



Figure A.3: LCD display showing STANDBY state — Room Empty



Figure A.4: LCD display showing ACTIVE state with countdown timer

A.3 Prototyping and Testing Phase

The following photographs document the prototyping and testing stages of the project, including breadboard assembly, component testing, and troubleshooting sessions.

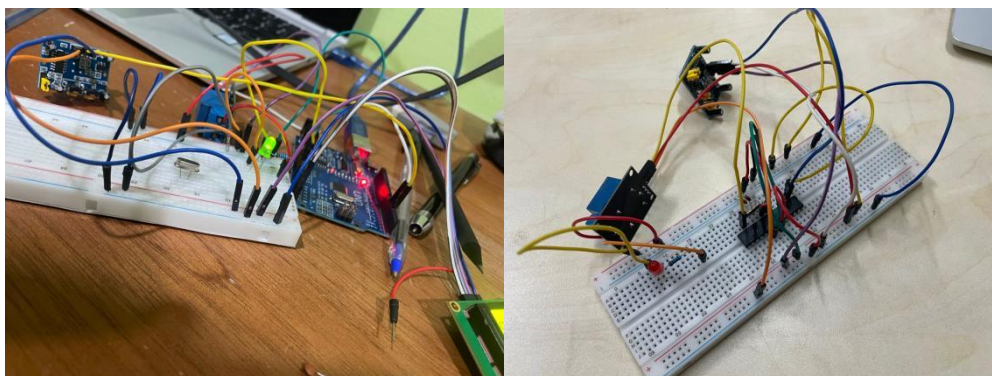


Figure A.5: Breadboard prototype during initial testing phase

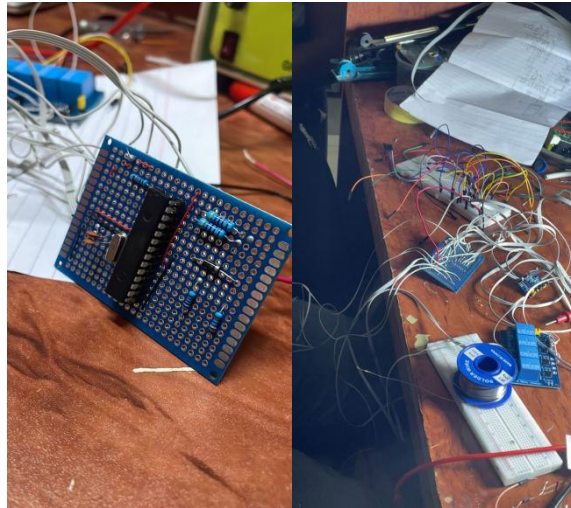


Figure A.6: Component testing and circuit wiring in progress